

Scalability & Future Plan

Date	29 September 2026
Team ID	UNM1659GenieBirds
Project Name	EduGenie – AI-Powered Learning Assistant
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current EduGenie solution can be scaled to handle larger user bases, increased data loads, and extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	AI response quality depends on the clarity and completeness of the user's input.	Vague questions may produce less targeted explanations or learning content.	High
2	The current prototype is designed for a limited user load and basic data storage.	Concurrent users and larger learning histories may increase response time and resource usage.	High
3	The current solution has limited personalization and advanced learner analytics.	Individual learning progress and recommendations can be improved with more user data and analytics.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Prototype supports a limited number of concurrent users.	Deploy using scalable cloud infrastructure, load balancing, and horizontal scaling for higher concurrent usage.
2	Data Storage	Learning data and interaction history are stored using a basic application-level storage approach.	Use a scalable database with indexing, backups, and structured storage for user profiles, learning history, and quiz data.
3	Performance	AI requests are processed through the current application workflow.	Introduce caching, asynchronous processing, optimized API calls, and monitoring to reduce latency as usage grows.
4	Security	Basic application protections are used in the prototype.	Add secure authentication, authorization, encrypted data handling, input validation, logging, and regular security reviews.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Enhanced personalization and learner profiles	1–3 months	More relevant explanations, quizzes, summaries, and learning recommendations.
Phase 2	Advanced analytics and progress tracking	3–6 months	Better visibility into learner progress, strengths, and improvement areas.
Phase 3	Scalable cloud deployment and database optimization	6–9 months	Improved reliability and support for larger numbers of users.
Phase 4	Additional learning resources and multimodal support	9–12 months	Broader learning experiences through richer educational content and interaction modes.